MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, MODEM

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Modem Circuit Card Assembly: 9355760, used in the Mortar Ballistics Computer (MBC): M23.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

FSC 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-454	-	Electronic Equipment, General Requirements for
MIL-STD-1521	-	Technical Reviews and Audits for Systems, Equipments, and Computer Programs
MIL-STD-45662	-	Calibration System Requirements

(copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Goverment documents, drawinqs, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal.

DRAWINGS

9355760	Circuit Card Assembly, Modem
9379254	Software, Test Station-Modem CCA
11785850	Computer Set, Ballistics, Mortar: M23

PACKAGING DATA SHEETS

9355760	Packaging of Circuit Card Assembly, Modem: 9355760

(Copies of drawings and packaging data sheets required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item definition. The modem circuit card assembly provides a digital communication link for the Mortar Ballistics Computer to communicate with other compatible Army systems. As shown in Figure 1, the three major sections of the circuit card assembly include the FSK modem, message preprocessor, and 1/0 control with a status register. The assembly also includes the control circuitry for the LED indicators and audible alarm. The connector contacts are “listed in Table I.

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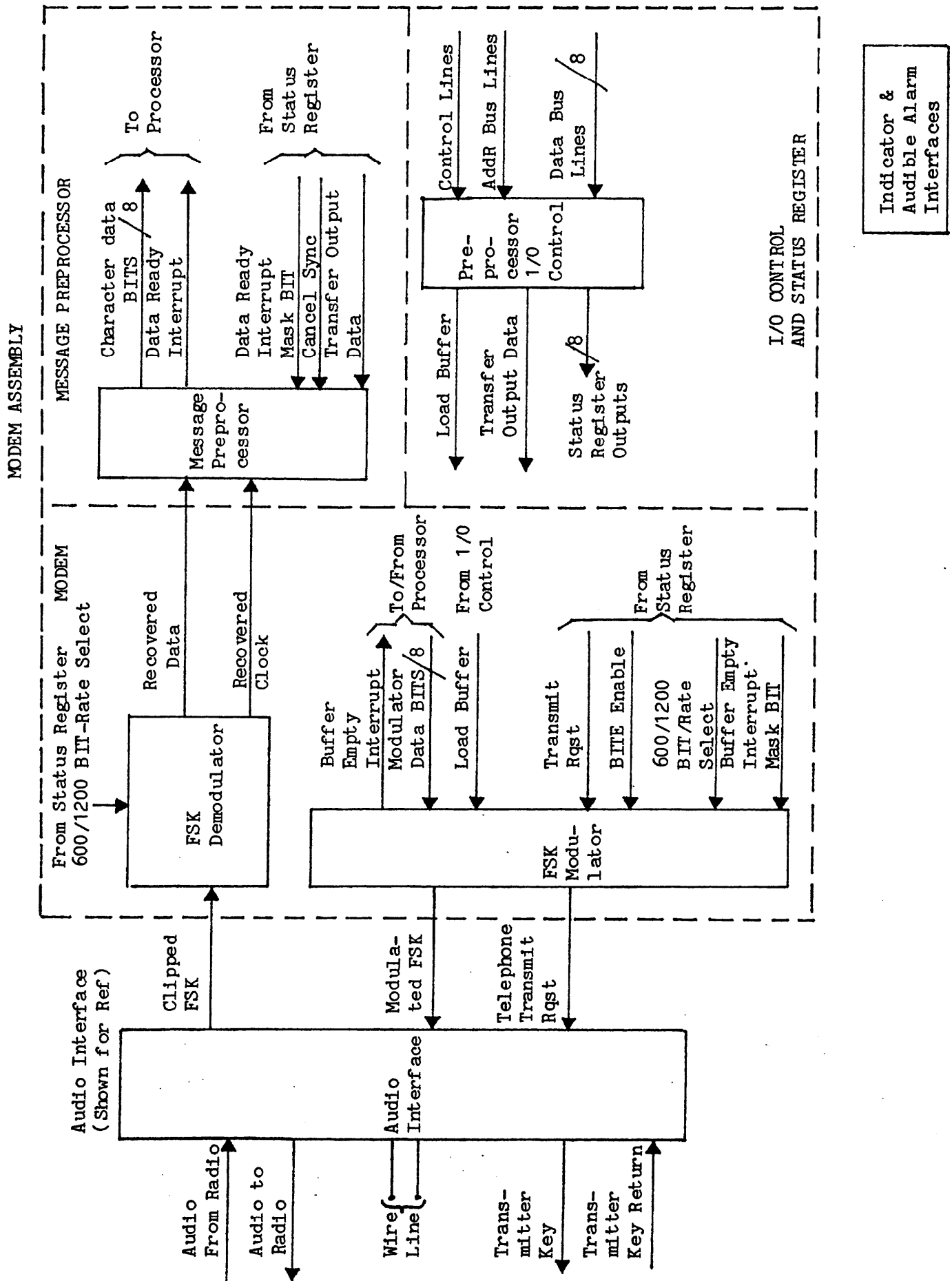


FIGURE 1. Modem block diagram.

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TABLE I. Modem circuit card assembly connector contracts.

Connector	Pin	Signal Name	Pin	Signal Name
P1	Al	+5V	B1	GND
P1	A2	AB00	B2	DB 0
P1	A3	AB01	B3	DB 1
P1	A4	AB02	B4	DB 2
Pl	A5	AB03	B5	DB3
P1	A6	AB04	B6	DB4
P1	A7		B7	DB 5
P1	A8		B8	DB 6
P1	A9		B9	DB7
P1	A10		B10	RD*
P1	All		Bll	WT*
P1	A12		B12	I0/M*
PI	Al 3		B13	
P1	A14		B14	
P1	Al 5		B15	Spare
P1	A16		B16	LE
P1	Al 7		B17	
P1	A18	+5V	B18	GND
P1	A19	CFSK	B19	
P1	A20	WXMT	B20	WIREN
P1	A21	MFSK	B21	Spare
P1	A22	MODINT*	1322	XMTKY
P1	A23	DEMODINT*	B23	XMTKYR
P1	A24	MC600	B24	MC600
P1	A25		B25	
P1	A26	DACLK*	B26	DCALK *
P1	A27	ROMICLK*	B27	ROMICLK*
P1	A28	ROM2CLK*	B28	ROM2CLK*
P1	A29	RCDATA	B29	RCDATA
P1	A30	RCCLK	B30	RCCLK
P1	A31	DCCFSK	B31	DCCFSK
P1	A32	MC19K	B32	MC19K
P1	A33	+5V	B33	GND
P1	A34	MBRST*	B34	614.4KHz
PI	A35	MCCLR	B35	5HZ
P1	A36	SYNCRST	B36	2Hz
P1	A37	DAENSET*	B37	lHz
P1	A38	DARST*	B38	2SEC
P1	A39	SAMREGRST*	B39	3SEC
P1	A40	DMODRST*	B40	6SEC
P1	a41	LDCNT40*	B41	LEDDC
P1	A42	RSTRDBC*	B42	DSPON*
P1	A43	Spare	B43	BATLOW
P1	A44	Spare	B44	Spare
P1	A45	SEQLED	B45	ONLED
P1	A46	BATLED	B46	MSGLED

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TABLE I. Modem circuit card assembly connector contacts.-Continued

P1	. A47	MC9.6K	B47	MSGLED
P1	A48	MC2400	B48	MC2400
P1	A49	MC1200	B49	MC1200
P1	A50	-5V	B50	GND

3.2 Performance. The performance of the Modem circuit card assembly shall be in accordance with the following requirements.

3.2.1 Functional. The modem circuit card assembly shall be attached to the Computer Automation CA8200 tester or equivalent, using the connections as specified in Table IV and pass the test patterns induced by Test Station Software-Modem CCA: 9379254.

3.2.2 Physical characteristics. The physical characteristics shall be as prescribed by the drawings, specifications and QAP's.

3.2.3 Reliability. When specified in the contract, a special sample shall be subjected to reliability testing. The manufacturing and production techniques emloyed in the fabrication of the Modem circuit card assembly shall not degrade the inherent reliability. The inherent reliability shall not be less than 84,422 hours Mean Time Between Failures (MTBF). (See 4 and 6)

3.3 Environmental. The modem circuit card assembly shall meet the requirements specified herein unless otherwise specified.

3.3.1 Room ambient temperature. The requirements of this specification shall be met with the modem assembly thermally stabilized at room ambient temperature of +160c to +32°C (+60° to +90°F), unless otherwise specified in the requirements paragraph.

3.3.2 Environmental stress screen (ESS) temperature. The modem circuit card assembly shall meet the requirements of 3.2.1 after having been exposed to a minimum of ten-temperature cycles at temperatures of -57°C and +71°C (-71°F and +160°F).

3.3.3 Environmental stress screen (ESS) vibration. The modem circuit card assembly shall meet the requirements of 3.2.1 after having been subjected to a random or pseudorandom vibration which includes frequency content from 100 Ilz to 1000 13z and up to a 3g RMS in two axis for a period of 10 minutes plus or minues 15 seconds.

3.4 Fabrication. The modem circuit card assembly shall be manufactured in accordance with drawing 9355760.

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3.5 First article. When specified in the contract, a sample shall be subjected to first article inspection (See 4 and 6).

3.6 workmanship. "The workmanship requirements of MIL-F-13926 and MIL-STD-454 shall be met. All components shall be fitted property and firmly in their respective positions. All electrical connections shall be mechanically secure and electrically sound. 'There shall be no evidence of corrosion, burrs, roughness, sharp edges or other defects which could adversely affect its use or suitability for the purpose intended.

3.7 Materials. " Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of a11 inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. "The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. "The component and subassembly inspection requirements of MIL-F-13926 and MIL-STD-454 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as "listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. "The sample for first article approval test shall consist of three (3) modem circuit card assemblies plus three (3) each of all items covered by QAP. The three modem circuit card assemblies shall be tested as specified herein for all the requirements of this specification. The items covered by QAP shall be tested, as specified thereon, for all characteristics delineated on the QAP. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts amd materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

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4.2.1 Government testing. when the Government is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements-of the contract. The sample and a record of this inspection, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractor's inspections.

4.3 Inspection provisions.

4.3.1 submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "Submission of Product" provision of MIL-sTD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected In accordance with the inspection provisions contained in Quality Assurance Provisions {QAP's) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4 Sampling.

4.4.1 Environmental\performance. The modem circuit card assembly shall be inspected to the requirements in Table II.

TABLE II. Classification of defects.

No.	Characteristics	Requirement	Test Procedure
100% Inspection			
1.	Functional test	3.2.1	4.6.3
2.	ESS temperature	3.3.2	4.6.1
3.	ESS vibration	3.3.3	4.6.2
MAJOR: AQL 1.0% Defective			
101.	Workmanship	3.6	4.6.4

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4.4.2 Special sampling. One modem circuit card assembly shall be selected at random as a special sample from each lot or each 100 produced which ever occurs first. The sample shall meet the requirements and tests in Table III.

TABLE XII. Classification of defects.

No,	Characteristics	Requirement	Test Procedure
201.	Fabrication	3.4	Applicable Drawing/ Visual
202.	Materials	3.7	Applicable Drawing/ Visual

NOTE: The tests in Table III shall be conducted at room ambient temperature (as specified in 3.3.1).

4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 contractor furnished inspection equipment.

4.5.2.1 Government design. All inspection equipment specified by drawing number in specification or QAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data package List (TDPL).

4.5.2.2 Contractor design. In the absence of Government inspection equipment design, the contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and “Test Facilities” requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the

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prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment desire requirements.

4.5.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test facilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F (1.4°C). The temperature gradient across the cross-sectional area occupied by-the test item shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radient heat, if used, will not fall directly on the test item. Thermocouples or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

4.5.2.3.2 Fabrication, setup and calibration. Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instruction shall be prepared and submitted to the procuring activity.

4.6 Test methods and procedures. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified tests. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

4.6.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The modem circuit card assembly shall be placed in the test chamber and the temperature of the chamber reduced 10°C (18°F') per minute plus or minus l°C (1.8°F) from room ambient temperature (as specified in 3.3.1) to -57°C (-71°F). The assembly shall be maintained at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure, the temperature of the

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chamber shall be raised 10°C (18°F) per minute plus or minus 1°C (1.8°F) to +71°F (+160°C). The assembly shall remain at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure the temperature of the chamber shall be reduced 10°C (18°F) per minute plus or minus 1°C (1.8°F) to room ambient temperature (as specified in 3.3.1). The above procedure shall be conducted a minimum of ten times. upon completion of the cycles the assembly shall then be removed from the chamber and subjected to the test in 4.6.3. It shall meet the requirements of 3.2.1.

4.6.2 ESS vibration. Conduct the vibration test with the modem circuit card assembly positioned on an appropriate adapter affixed to a vibrating machine that is capable of providing the amplitudes and frequencies specified in 3.3.3. Unless otherwuse specified, the test equipment shall be in accordance with the "Test Facilities" requirements of Mil-F-13926. The applied motion shall be in two separate planes for the time period specified. Upon completion of the vibration sweep the assembly shall be removed from the vibration adapter and shall be subjected to the tests in 4.6.3.It shall meet the requirements of 3,2.1.1 and 3.2.1.7 inclusive.

4.6.3 Functional tests. This test is to be conducted utilizing the Computer Automation CA8200 or equivalent. Use of an equivalent computer must have prior approval-of the procuring activity. The display/processor circuit card assembly shall be attached to the CA8200 utilizing appropriate fixtures and/or cables and connected as specified in Table IV. The assembly shall be tested in accordance with the test patterns contained on the Software Test Station-Modem CCA: 9379254, provided by the procuring activity. The assembly shall meet the requirements of 3.2.1.

TABLE IV. Modem circuit card assembly test fixture connector contacts.

Display/Processor		Connect to	CA8200
Connector	Pin		Pin
P1	Al		63
P1	A2		2
P1	A3		3
P1	A4		4
P1	A5		5
P1	A6		6
P1	A18		63
P1	A20		18
P1	A21		160
P1	A22		20

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TABLE IV. Modern circuit card assembly test fixture connector contacts.-Continued

P1	A23	21
P1	A24	22
Pl	A26	24
P1	A27	25
P1	A28	26
P1	A29	27
P1	A30	28
P1	A31	161
P1	A32	32
P1	A33	63
P1	A34	34
P1	A35	35
Pl	A36	36
P1	A37	37
P1	A38	38
P1	A39	39
P1	A40	40
P1	A41	41
P1	A42	42
P1	A47	47
P1	A48	48
P1	A49	49
P1	A50	GND
P1	B1	414
P1	B2	52
P1	B3	53
P1	B4	54
P1	B5	55
P1	B6	56
P1	B7	57
P1	B8	58
PI	B9	59
P1	B10	60
P1	Bll	61
P1	B12	64
P1	B16	68
P1	B18	446
P1	B20	72
P1	B22	74
P1	B24	76
P1	B26	78
P1	B27	79
P1	B28	80
P1	B29	81
P1	B30	82
P1	B31	83
P1	B32	84

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TABLE IV. Modem circuit card assembly test fixture connector contacts.-Continued

P1	B33	GND
P1	B34	86
P1	B35	87
P1	B36	88
P1	B37	89
P1	B38	90
P1	B39	91
P1	B40	92
P1	B41	93
P1	B42	96
P1	B43	97
P1	B47	101
P1	B48	102
P1	B49	103
P1	B50	GND
TP	1	105
TP	2	106
TP	3	107
TP	4	108
TP	5	109
TP	6	110
TP	7	111
TP	8	112
TP	9	113
TP	10	114
TP	11	115
TP	12	116
TP	13	117
TP	14	118
TP	15	119
TP	16	120

4.6.4 Workmanship. The modem circuit card assembly shall be subjected to a visual and tactile “examination for evidence of faulty workmanship. The assembly shall conform to the requirements of 3.6.

4.7 Reliability testing. The requirement for reliability test approval and the responsibility (Government or contractor) for reliability testing shall be as-specified in the contract. The sample for reliability test shall consist of 3 modem circuit card assemblies. The samples shall be manufactured in the same manner, using the same materidls, equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

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4.7.1 Government testing. When the Government is responsible for conducting reliability tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4,7.2 Contractor testing. When the contractor is responsible for conducting reliability tests, the sample shall be inspected and tested by-the contractor for all the requirements of the contract. The sample and a record of these inspections and tests, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractors reliability tests and inspections.

5. PREPARATION FOR DELiVERY

5.1 Delivery requirements. Handling, packaging, packing, shipping, and marking of the modem circuit card assembly, 9355760 shall be in accordance with Packaging Data Sheet 9355760. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The modem circuit card assembly described herein is intended for use in the MBC. See drawing 11785850.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for first article testing.

c. Provisions for reliability testing.

d. Applicable Packaging Data Sheet.

e. Applicable-levels of preservation, packing, and marking (see 5.1).

f. Applicable stock numbers.

Custodian:
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Preparing activity:
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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